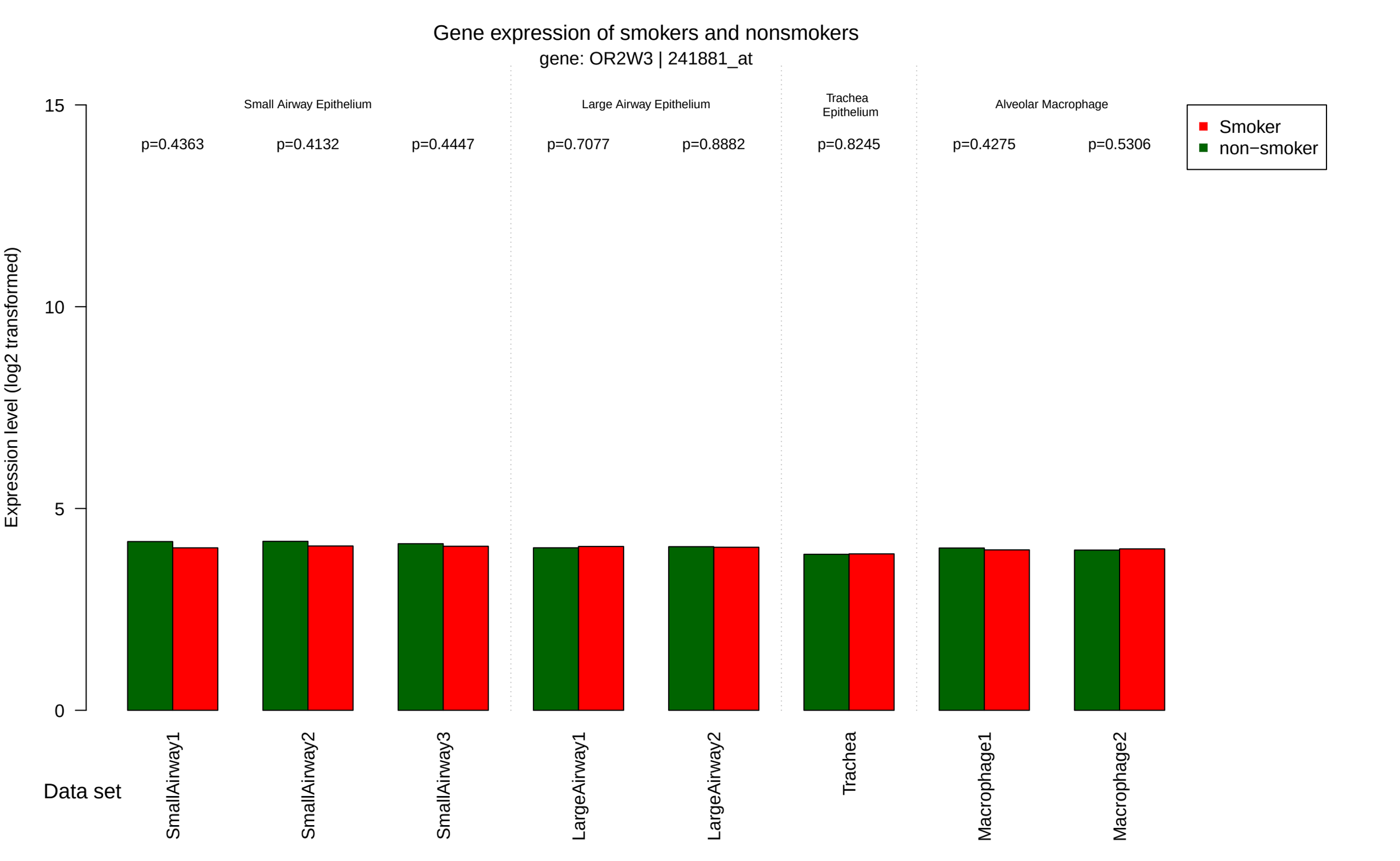
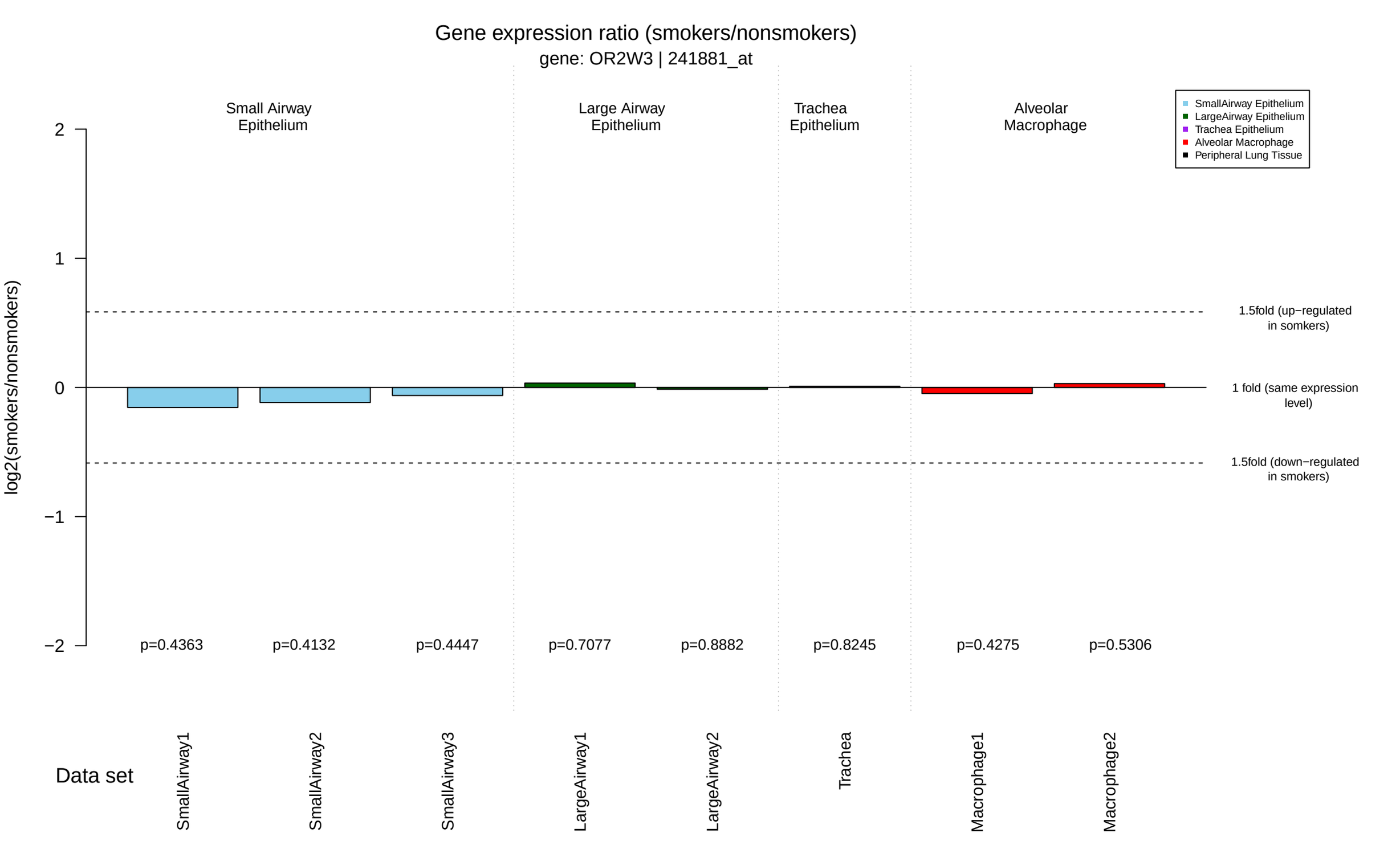
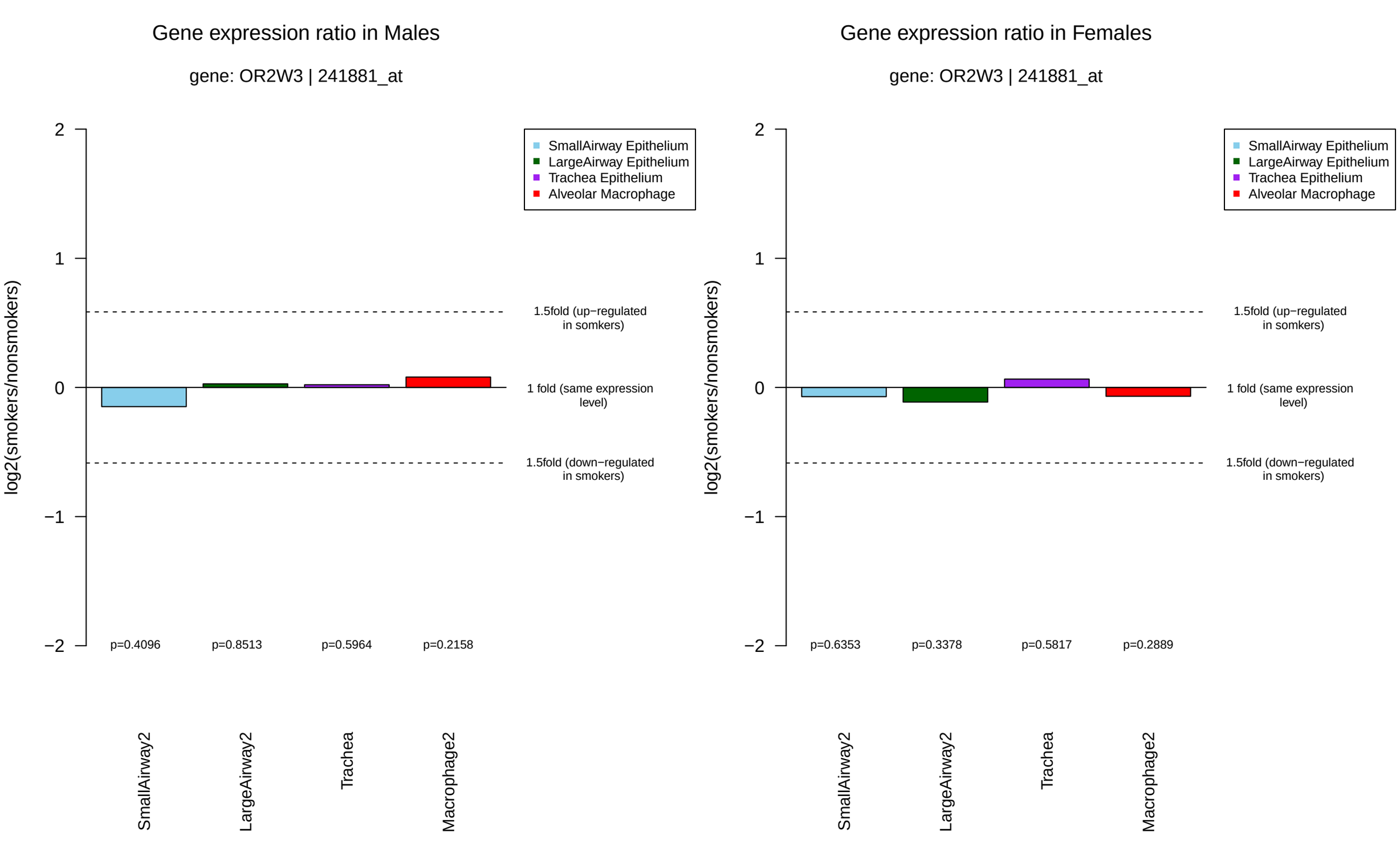
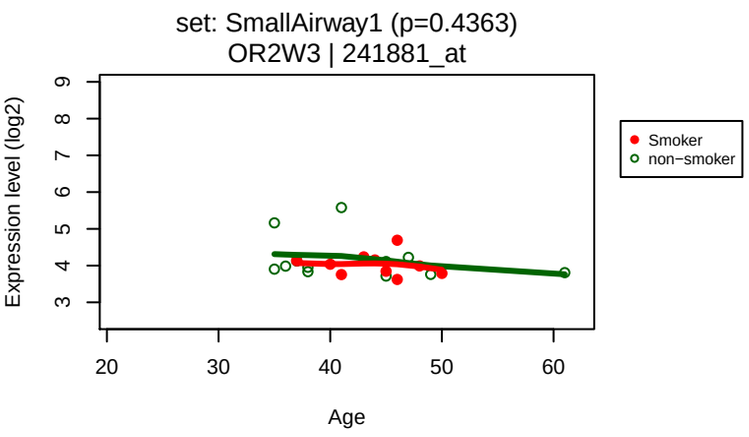
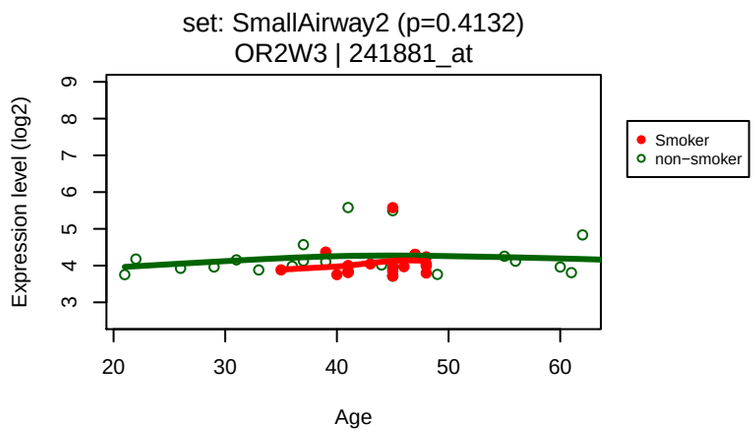
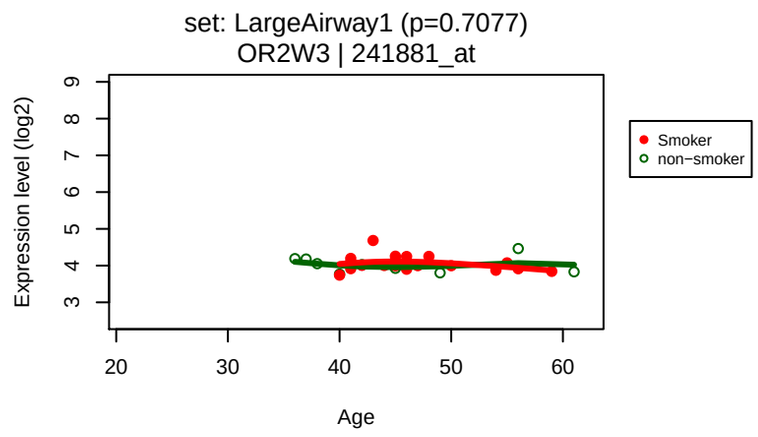

Gene expression of smokers and nonsmokers
gene: OR2W3 | 241881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4363
p=0.4132
p=0.4447
p=0.7077
p=0.8882
p=0.8245
p=0.4275
p=0.5306
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

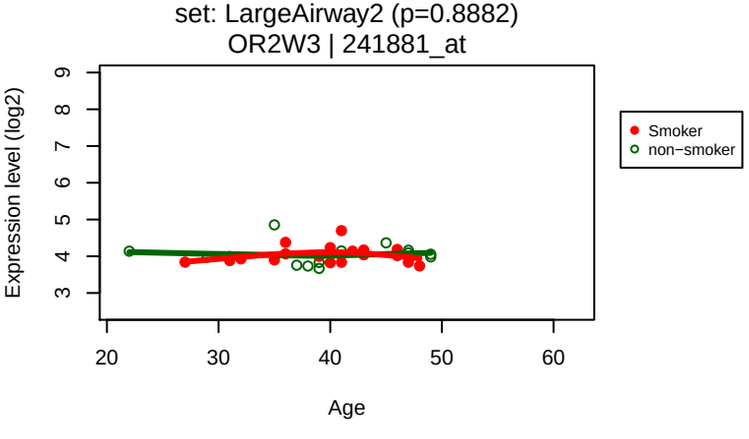

Gene expression ratio (smokers/nonsmokers)
gene: OR2W3 | 241881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4363
p=0.4132
p=0.4447
p=0.7077
p=0.8882
p=0.8245
p=0.4275
p=0.5306
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

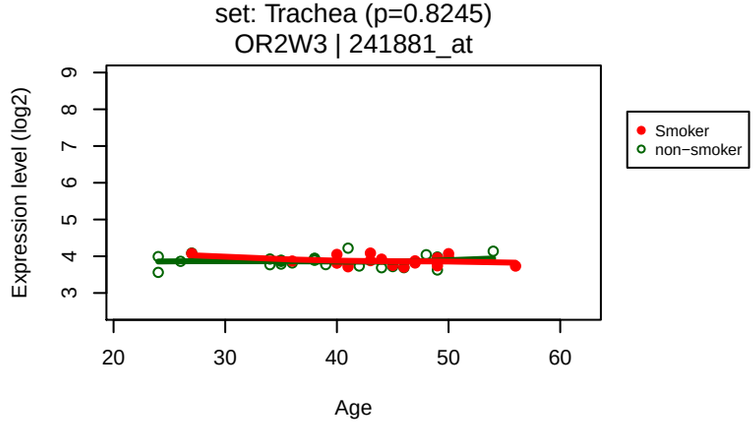

Gene expression ratio in Males
gene: OR2W3 | 241881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4096
p=0.8513
p=0.5964
p=0.2158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2W3 | 241881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6353
p=0.3378
p=0.5817
p=0.2889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

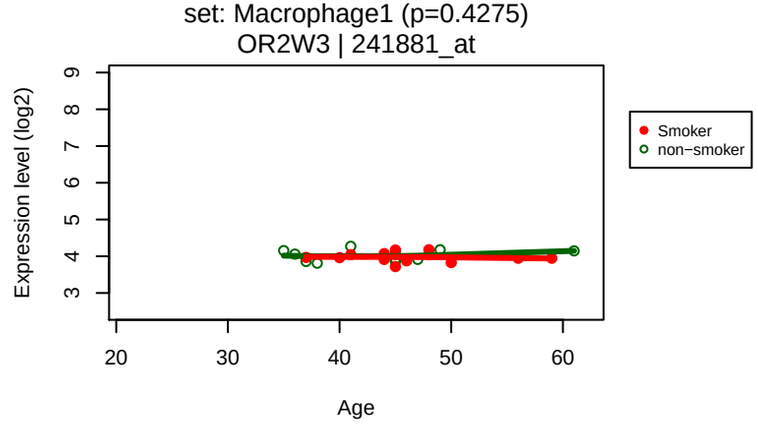

set: SmallAirway1 (p=0.4363)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker

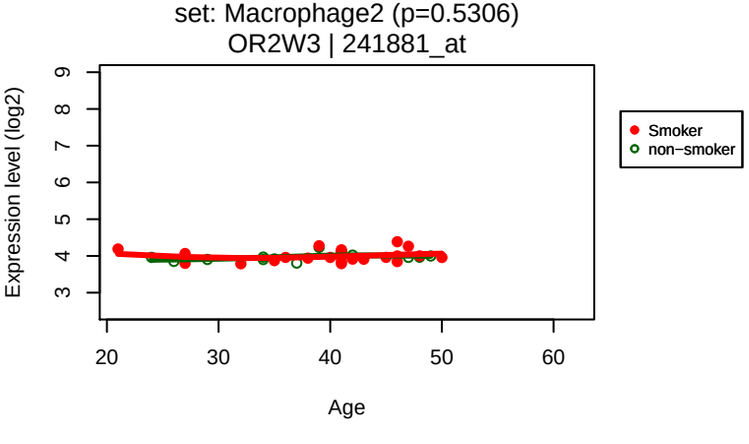

set: SmallAirway2 (p=0.4132)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker

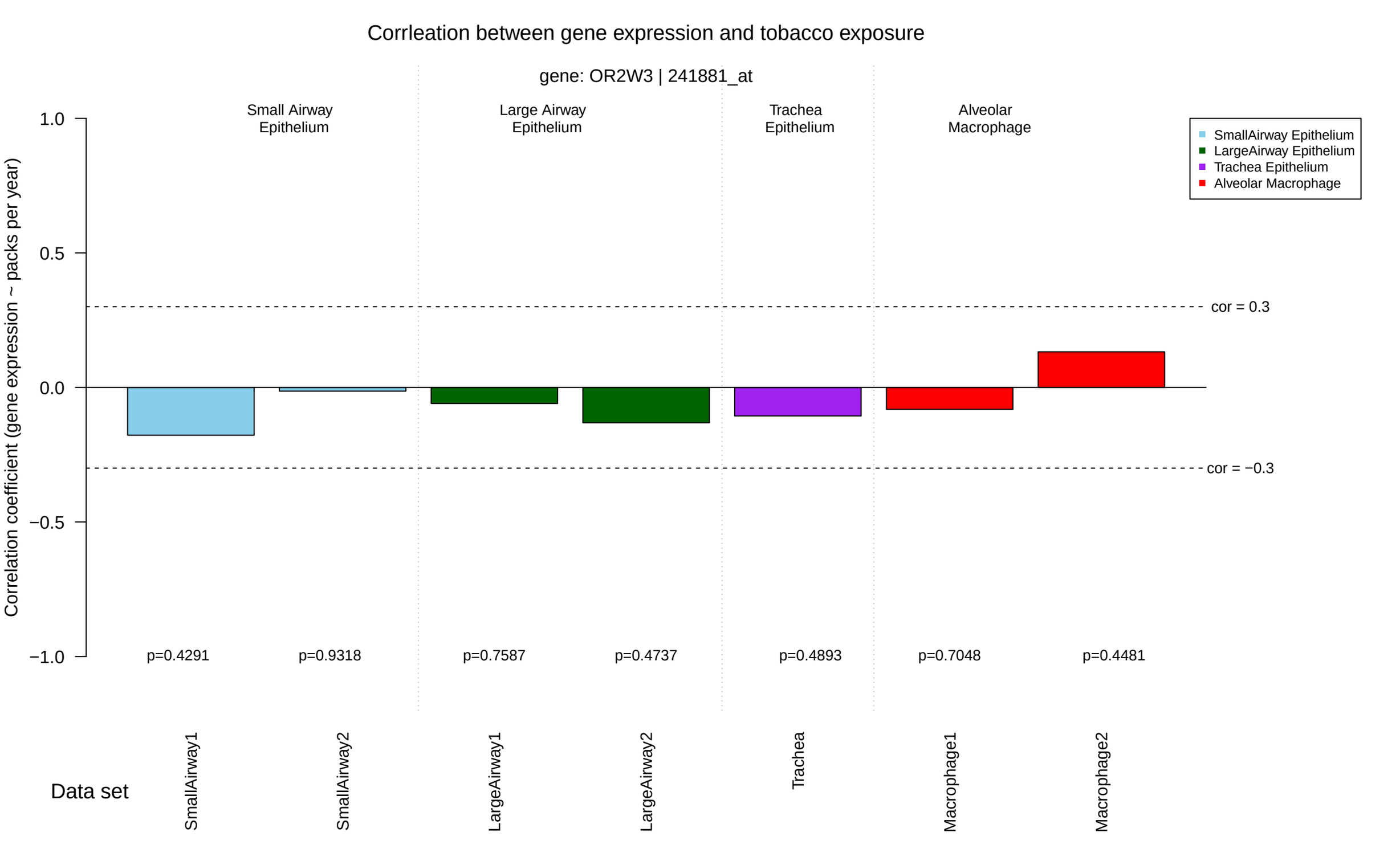

set: LargeAirway1 (p=0.7077)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8882)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8245)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4275)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5306)
OR2W3 | 241881_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2W3 | 241881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4291
p=0.9318
p=0.7587
p=0.4737
p=0.4893
p=0.7048
p=0.4481
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2